United States Patent	12401048
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Aslam; Muhammad Kashif et al.

Metal-carbon dioxide (CO.SUB.2.) battery cell

Abstract

The present disclosure is directed to a zero-gap flow-type metal-Carbon dioxide CO.sub.2 battery cell. The battery cell includes an electrolyte chamber allowing flow of electrolyte and a metal electrode configured as anode, being accommodated in the electrolyte chamber and defined with a plurality of apertures. Further, the battery cell includes a CO.sub.2 chamber allowing flow of CO.sub.2 and a gas diffusion electrode (GDE) configured as cathode and accommodated in the CO.sub.2 chamber. The GDE is defined with a first side and a second side, which is coated with catalyst layer. The catalyst layer abuts a membrane on the metal electrode such that, the electrolyte from the electrolyte chamber flows through the plurality of apertures, penetrates through the membrane and contacts the catalyst layer. Further, CO.sub.2 from the CO.sub.2 chamber contacts the second side of the gas diffusion electrode to diffuse through the catalyst and towards the metal electrode.

Inventors: **Aslam; Muhammad Kashif (Al Ain, AE), Al Marzouqi; Ali (Al Ain, AE)**

Applicant: **United Arab Emirates University** (Al Ain, AE)

Family ID: **1000008447370**

Assignee: **UNITED ARAB EMIRATES UNIVERSITY (Al-Ain, AE)**

Appl. No.: **19/032872**

Filed: **January 21, 2025**

Publication Classification

Int. Cl.: **H01M4/86** (20060101); **H01M4/90** (20060101); **H01M12/02** (20060101); **H01M12/08** (20060101)

U.S. Cl.:

CPC **H01M4/8657** (20130101); **H01M4/8605** (20130101); **H01M4/9008** (20130101); **H01M12/02** (20130101); **H01M12/08** (20130101); H01M2300/0014 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2024/0141517	12/2023	Cowan	N/A	H01M 4/8807
2024/0222744	12/2023	Valsecchi	N/A	H01M 4/42

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2022129895	12/2021	WO	B01J 23/34
WO-2022238584	12/2021	WO	H01M 12/08
WO-2024086935	12/2023	WO	N/A

Primary Examiner: Dove; Tracy M

Attorney, Agent or Firm: MH2 Technology Law Group, LLP

Background/Summary

TECHNOLOGICAL FIELD

(1) The present disclosure in general relates to the field of generation and distribution of electric power by batteries. The present disclosure is further directed towards a zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery cell and a battery module thereof.

BACKGROUND

(2) The information in this section merely provides background information related to the present disclosure and may not constitute prior art(s) for the present disclosure.

(3) Emission of greenhouse gases is continuously increasing with industrialization and carbon dioxide [CO.sub.2] amounts to one of the largest proportions of greenhouse gases. Considering this, various systems have evolved which utilize CO.sub.2 to produce energy and high value hydrocarbons. On such system is Metal-CO.sub.2 battery, which utilizes CO.sub.2 for energy regeneration. In Metal-CO.sub.2 batteries, a metal anode oxidizes and releases electrons, which are then transferred through an external circuit to a cathode to reduce CO.sub.2. Conventional Metal-CO.sub.2 batteries operate with non-aqueous electrolyte, which pose limitations such as poor rate capability, high voltage gap and short cycle life due to accumulation of solid discharge residue on surface of electrodes or catalyst of said battery. Further, in the conventional batteries, proton-coupled electron transfer (PCET) reaction is a challenge, due to non-aqueous electrolyte, which tends to inhibit reaction of CO.sub.2 to hydrocarbons. In addition, structural configuration of the conventional Metal-CO.sub.2 battery not only makes it bulky, but also increases ionic resistance during transport of ions between the metal anode and the cathode. This leads to slower ion transport, which in-turn tends to limit battery cell reaction kinetics, thus affecting performance of the Metal-CO.sub.2 battery.

(4) Present disclosure is directed to overcome one or more limitations stated above or any other limitations associated with the known arts.
(5) General Description
(6) One or more shortcomings of the prior art are overcome by a device and a system as claimed and additional advantages are provided through the present disclosure. Additional features and advantages are realized through the techniques of the present disclosure. Other embodiments, implementations and aspects of the disclosure are described in detail herein and are considered a part of the claimed disclosure.
(7) A first aspect of the disclosure concerns a zero-gap flow-type metal-Carbon dioxide (CO_2) battery cell. The battery cell includes an electrolyte chamber configured to allow flow of electrolyte and a metal electrode configured as an anode, where the metal electrode is accommodated in the electrolyte chamber and is defined with a plurality of apertures. Further, the battery cell includes a Carbon dioxide (CO_2) chamber configured to allow flow of Carbon dioxide (CO_2) and a gas diffusion electrode (GDE) configured as a cathode, where the gas diffusion electrode (GDE) is accommodated in the Carbon dioxide (CO_2) chamber. The gas diffusion electrode (GDE) is defined with a first side and a second side, opposite to the first side, and the second side is coated with a catalyst layer. The catalyst layer abuts a membrane on the metal electrode such that, the electrolyte from the electrolyte chamber flows through the plurality of apertures in the metal electrode **105**, penetrates through the membrane and contacts the catalyst layer. Further, Carbon dioxide (CO_2) from the Carbon dioxide (CO_2) chamber contacts the second side of the gas diffusion electrode to diffuse through the catalyst and towards the metal electrode.
(8) According to a configuration of the first aspect, the gas diffusion electrode comprises an intermediate layer sandwiched between the first side of the gas diffusion electrode and the catalyst layer. The intermediate layer is formed using a mixture of Polytetrafluoroethylene (PTFE) and carbon nanospheres.
(9) According to a configuration of the first aspect, the metal electrode is a Zinc electrode and the gas diffusion electrode is made of carbon paper.
(10) According to a configuration of the first aspect, the electrolyte is 6M KOH+0.02M $Zn(CH_3COO)_2$ and the membrane is made of a glass fibre.
(11) According to a configuration of the first aspect, the catalyst layer is formed of Bismuth metal-organic framework (Bi-MOF).
(12) According to a configuration of the first aspect, the electrolyte chamber is defined with a first inlet and a first outlet. The electrolyte enters through the first inlet and discharges through the first outlet.
(13) According to a configuration of the first aspect, the electrolyte chamber is defined with a first cavity along an inner periphery to define a flow path for the flow of the electrolyte from the first inlet and towards the first outlet.
(14) According to a configuration of the first aspect, the battery module includes a first gasket surrounding the metal electrode and abutting to inner walls of the first cavity.
(15) According to a configuration of the first aspect, the Carbon dioxide (CO_2) chamber is defined with a second inlet and a second outlet. Carbon dioxide (CO_2) enters through the second inlet and discharges through the second outlet.
(16) According to a configuration of the first aspect, the Carbon dioxide (CO_2) chamber is defined with a second cavity along an inner periphery to define a flow path for the flow of Carbon dioxide (CO_2).
(17) According to a configuration of the first aspect, the battery module includes a second gasket surrounding the gas diffusion electrode and abutting inner walls of the second cavity.
(18) According to a configuration of the first aspect, the battery module includes at least one electric terminal extending from each of the metal electrode and the gas diffusion electrode to conduct electrons from the metal electrode towards the gas diffusion electrode.

(19) A second aspect of the disclosure concerns a zero-gap flow-type metal-Carbon dioxide battery module. The battery module includes a plurality of zero-gap flow-type metal-Carbon dioxide battery cells stacked within the casing in a defined orientation. Each of the plurality of battery cells includes an electrolyte chamber configured to allow flow of electrolyte and a metal electrode configured as an anode, where the metal electrode is accommodated in the electrolyte chamber and is defined with a plurality of apertures. Further, the battery cell includes a Carbon dioxide (CO.sub.2) chamber configured to allow flow of Carbon dioxide (CO.sub.2) and a gas diffusion electrode (GDE) configured as a cathode, where the gas diffusion electrode (GDE) is accommodated in the Carbon dioxide (CO.sub.2) chamber. The gas diffusion electrode (GDE) is defined with a first side and a second side, opposite to the first side, and the second side is coated with a catalyst layer. The catalyst layer abuts a membrane on the metal electrode such that, the electrolyte from the electrolyte chamber flows through the plurality of apertures in the metal electrode, penetrates through the membrane and contacts the catalyst layer. Further, Carbon dioxide (CO.sub.2) from the Carbon dioxide (CO.sub.2) chamber contacts the second side of the gas diffusion electrode to diffuse through the catalyst and towards the metal electrode.
(20) According to a configuration of the second aspect, the battery module includes a pump adapted to circulate the electrolyte and Carbon dioxide to each of the plurality of zero-gap flow-type metal-Carbon dioxide battery cells stacked within the casing.
(21) The foregoing description is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments 1 and features described above, further aspects, embodiments 1, and features will become apparent by reference to the drawings and the following description.
Embodiment 1
(22) The present disclosure also encompasses embodiment 1 as defined in the following numbered phrases. It should be noted that these numbered embodiments 1 are intended to add to this disclosure and are not intended in any way to be limiting. 1. A zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery cell, the battery cell comprising: an electrolyte chamber configured to allow flow of electrolyte; a metal electrode configured as an anode, the metal electrode is accommodated in the electrolyte chamber and is defined with a plurality of apertures; a Carbon dioxide (CO.sub.2) chamber configured to allow flow of Carbon dioxide (CO.sub.2); and a gas diffusion electrode (GDE) configured as a cathode, the gas diffusion electrode (GDE) is accommodated in the Carbon dioxide (CO.sub.2) chamber, the gas diffusion electrode (GDE) is defined with a first side and a second side, opposite to the first side, and the second side is coated with a catalyst layer, wherein, the catalyst layer abuts a membrane on the metal electrode such that, the electrolyte from the electrolyte chamber flows through the plurality of apertures in the metal electrode, penetrates through the membrane and contacts the catalyst later, and Carbon dioxide (CO.sub.2) from the Carbon dioxide (CO.sub.2) chamber contacts the second side of the gas diffusion electrode to diffuse through the catalyst and towards the metal electrode. 2. The battery cell of embodiment 1, wherein the gas diffusion electrode comprises an intermediate layer sandwiched between the first side of the gas diffusion electrode and the catalyst layer, wherein the layer is formed using a mixture of Polytetrafluoroethylene (PTFE) and carbon nanospheres. 3. The battery cell **100** of embodiment 1, wherein the metal electrode is a Zinc electrode. 4. The battery cell of embodiment 1, wherein the gas diffusion electrode is made of carbon paper. 5. The battery cell of embodiment 1, wherein the electrolyte is 6M KOH+0.02M Zn(CH.sub.3COO).sub.2). 6. The battery cell of embodiment 1, wherein the membrane is made of a glass fibre. 7. The battery cell of embodiment 1, wherein the catalyst layer is formed of Bismuth metal-organic framework (Bi-MOF). 8. The battery cell of embodiment 1, wherein the electrolyte chamber is defined with a first inlet and a first outlet, wherein the electrolyte enters through the first inlet and discharges through the first outlet. 9. The battery cell of embodiments 1 and 8, wherein the electrolyte chamber is defined with a first cavity along an inner periphery to define a flow path for the flow of

the electrolyte from the first inlet and towards the first outlet. 10. The battery cell of embodiments 1 and 9, comprising a first gasket surrounding the metal electrode and abutting to inner walls of the first cavity. 11. The battery cell of embodiment 1, wherein the Carbon dioxide (CO.sub.2) chamber is defined with a second inlet and a second outlet, wherein Carbon dioxide (CO.sub.2) enters through the second inlet and discharges through the second outlet. 12. The battery cell of embodiment 1, wherein the Carbon dioxide (CO.sub.2) chamber is defined with a second cavity along an inner periphery to define a flow path for the flow of Carbon dioxide (CO.sub.2). 13. The battery cell of embodiment 1 and 12, comprising a second gasket surrounding the gas diffusion electrode and abutting inner walls of the second cavity. 14. The battery cell of embodiment 1, comprising at least one electric terminal extending from each of the metal electrode and the gas diffusion electrode to conduct electrons from the metal electrode towards the gas diffusion electrode. 15. A zero-gap flow-type metal-Carbon dioxide battery module, the module comprising: a casing; and a plurality of zero-gap flow-type metal-Carbon dioxide battery cells stacked within the casing in a defined orientation, each of the plurality of zero-gap flow-type metal-Carbon dioxide cells comprises: an electrolyte chamber configured to allow flow of electrolyte; a metal electrode configured as an anode, the metal electrode is accommodated in the electrolyte chamber and is defined with a plurality of apertures; a Carbon dioxide (CO.sub.2) chamber configured to allow flow of Carbon dioxide (CO.sub.2); and a gas diffusion electrode (GDE) configured as a cathode, the gas diffusion electrode (GDE) is accommodated in the Carbon dioxide (CO.sub.2) chamber, the gas diffusion electrode (GDE) is defined with a first side and a second side, opposite to the first side, and the second side is coated with a catalyst layer, wherein, the catalyst layer abuts a membrane on the metal electrode such that, the electrolyte from the electrolyte chamber flows through the plurality of apertures in the metal electrode, penetrates through the membrane and contacts the catalyst later, and Carbon dioxide (CO.sub.2) from the Carbon dioxide (CO.sub.2) chamber contacts the second side of the gas diffusion electrode to diffuse through the catalyst and towards the metal electrode. 16. The battery module of embodiment 15, comprising a pump adapted to circulate the electrolyte and Carbon dioxide to each of the plurality of zero-gap flow-type metal-Carbon dioxide battery cells stacked within the casing.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to better understand the subject matter that is disclosed herein and to exemplify how it may be carried out in practice, embodiments 1 will now be described, by way of non-limiting examples only, with reference to the accompanying drawings, in which:

(2) FIG. **1** illustrates an explode view of a zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery cell, according to an example of the present disclosure.

(3) FIG. **2** illustrates an assembled perspective view of the zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery cell of FIG. **1**.

(4) FIG. **3** illustrates a top sectional view of the zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery cell of FIG. **2**.

(5) FIG. **4** illustrates a side view of a gas diffusion electrode, according to an example of the present disclosure.

(6) FIG. **5** illustrates a schematic sectional view of a portion of the battery cell, depicting diffusion of CO.sub.2 from the CO.sub.2 chamber, according to an example of the present disclosure.

(7) FIG. **6** illustrates a schematic view of a zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery module, according to an example of the present disclosure.

(8) FIG. **7***a*-**7***c* depicts polarization curves obtained during operation of the battery cell, according to an example of the present disclosure.

DETAILED DESCRIPTION

(9) FIG. **1** illustrates an exploded view of a zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery cell **100** [hereinafter interchangeably referred as battery cell **100**], in accordance with an exemplary implementation of the present disclosure. The term 'flow-type" as used in the present disclosure refers to a feature of the battery cell **100** where, working fluids such as electrolyte and/or Carbon dioxide [hereinafter interchangeably referred as CO.sub.2] are continuously circulated into the battery cell **100** to generate, convert and/or distribute energy, and other byproducts. In an implementation, a plurality of battery cells **100** may be stacked in a defined orientation to form a zero-gap flow-type metal-Carbon dioxide (CO.sub.2) battery module **200** [hereinafter referred interchangeably as battery module **200**]. As seen in FIG. **6**, the battery module **200** includes a casing **201** inside which a plurality of battery cells **100** are stacked in the defined orientation. As an example, the defined orientation may be one of a vertical orientation or a horizontal orientation. The battery module **200** includes one or more reservoirs **202***a*, **202***b* for storing the working fluids and one or more pumps **203***a*, **203***b* associated with each of the one or more reservoirs **202***a*, **202***b*, respectively. The one or more pumps **203***a*, **203***b* are configured for circulating the working fluids into each of the plurality of battery modules **100** to facilitate operation of each of the plurality of battery modules **100** to generate energy and other by-products such as hydrocarbons. As an example, the one or more pumps **203***a*, **203***b* may be but not limiting to a peristaltic pump. In an illustrated embodiment, as seen in FIG. **6**, one reservoir **202***a* is adapted to store the electrolyte and the pump **203***a* is configured to circulate the electrolyte into each of the plurality of battery modules and likewise, one reservoir **202***b* is adapted to store CO.sub.2 and the pump **203***b* is configured to circulate the CO.sub.2 into each of the plurality of battery modules **100**. However, the same cannot be construed as a limitation, as individual reservoir may be associated with each of the battery modules **100** for supplying electrolyte and CO.sub.2 into the battery modules **100**. Referring to FIG. **1**, the battery cell **100** includes an electrolyte chamber **101** which is configured to allow flow of electrolyte. In an implementation, the electrolyte chamber **101** may be defined in any geometrical shape such as not limiting to square, rectangle and the like, depending on configuration of the battery cell **100** and its intended application. The electrolyte chamber **101** may be defined with a first cavity **102** on an inner surface **103**, i.e., the inner surface **103** is a surface of the electrolyte chamber **101** which is not exposed to surroundings of the battery cell **100** or to adjacent battery cell **100** in the battery module **200**, while an outer surface, opposite to the inner surface **103** of the electrolyte chamber **101** may be covered or selectively exposed to the surrounding or the adjacent battery cell **100** of the battery module **200**. In an implementation, the first cavity **102** extends proximal to a periphery and at a distance from sides of the electrolyte chamber **101**. The first cavity **102** may define a flow path for flow of the electrolyte during operation of the battery cell **100**. Further, the electrolyte chamber **101** is defined with a first inlet **117** and a first outlet **104**. The first inlet **117** may be defined proximal to a bottom end at one side of the electrolyte chamber **101** and the first outlet **104** may be defined proximal to a top end at another side, opposite to one side of the electrolyte chamber **101**. The electrolyte enters through the first inlet **117**, flows through the flow path and discharges out of the electrolyte chamber **101** through the first outlet **104**. As an example, the electrolyte may be including, but not limiting to, an aqueous 6M KOH+0.02M Zn(CH.sub.3COO).sub.2).

(10) Further referring to FIG. **1**, the battery cell **100** includes a metal electrode **105** which is configured to act as an anode. As an example, the metal electrode **105** may be including, but not limiting to, a Zinc electrode. The metal electrode **105** is accommodated in the electrolyte chamber **101** and is defined with a plurality of apertures **106**. In an implementation, one aperture of the plurality of apertures **106** is defined in proximity to one corner **107***a* of the metal electrode **105**, and other aperture of the plurality of apertures **106** is defined in proximity to other corner **107***b* which is diagonally opposite to the one corner **107***a* of the metal electrode **105**. In an implementation, the metal electrode **105** is accommodated in the electrolyte chamber **101** where the

metal electrode **105** covers the first cavity **102** such that, the electrolyte entering the first inlet **117** of the electrolyte chamber **101** flows through the first flow path and through the plurality of apertures **106** and discharges through the first outlet **104**. The plurality of apertures **106** enables effective electrolyte penetration through the metal electrode **105**, thus enhances overall reaction kinetics. In an implementation, the battery cell **100** includes a first gasket **108** which is positioned to surround the metal electrode **105** and abut a portion of the inner surface **103** of the electrolyte chamber **101**. The first gasket **108** forms a sealing between the metal electrode **105** and the first cavity **102** [i.e., the electrolyte chamber **101**], forming a leak proof flow path for the electrolyte.
(11) Further, the battery cell **100** includes a Carbon dioxide (CO.sub.2) chamber **109** which is configured to allow flow of Carbon dioxide (CO.sub.2). In an implementation, the Carbon dioxide (CO.sub.2) chamber **109** may be defined in any geometrical shape such as not limiting to square, rectangle and the like depending on the configuration of the battery cell **100**, and complements to shape of the electrolyte chamber **101**. The Carbon dioxide (CO.sub.2) chamber **109** may be defined with a second cavity **110** on an inner surface **118**, i.e., the inner surface **118** is a surface of the Carbon dioxide (CO.sub.2) chamber **109** which is not exposed to surroundings of the battery cell **100** or to adjacent battery cell **100** in the battery module **200**, while an outer surface, opposite to the inner surface **118** of the Carbon dioxide (CO.sub.2) chamber **109** may be covered or selectively exposed to the surrounding or the adjacent battery cell **100** of the battery module **200**. In an implementation, the second cavity **110** extends along a periphery and at a distance from sides of the Carbon dioxide (CO.sub.2) chamber **109**. The second cavity **110** may define a flow path for flow of CO.sub.2 during operation of the battery cell **100**. Further, the Carbon dioxide (CO.sub.2) chamber **109** is defined with a second inlet **112** and a second outlet **113**. The second inlet **112** may be defined in proximity to a bottom end at one side of the Carbon dioxide (CO.sub.2) chamber **109** and the second outlet **113** may be defined in proximity to a top end at another side, opposite to one side. The Carbon dioxide (CO.sub.2) enters through the second inlet **112**, flows through the flow path and discharges out of the Carbon dioxide (CO.sub.2) chamber **109**, through the second outlet **113**. In an implementation, the battery cell **100** includes a first gasket **108** which is positioned to surround the metal electrode **105** and abut a portion of the inner surface **103** of the electrolyte chamber **101**. The first gasket **108** forms a sealing between the metal electrode **105** and the first cavity **102** [i.e., the electrolyte chamber **101**], forming a leak proof flow path for the electrolyte.
(12) Referring again to FIG. **1**, the battery cell **100** includes a gas diffusion electrode **111** which may be configured to act as a cathode and is accommodated in the Carbon dioxide (CO.sub.2) chamber **109**. As an example, the gas diffusion electrode **111** (GDE) is made of material including, but not limiting to, a carbon paper. The gas diffusion electrode **111** (GDE) is defined with a first side **123** and a second side **124**, which is opposite to the first side **123**. The second side **124** is coated with a catalyst layer **114**, while the first side **123** may be configured to seat within the Carbon dioxide (CO.sub.2) chamber **109**. As an example, the catalyst layer **114** may be formed of material including, but not limiting to, Bismuth metal-organic framework (Bi-MOF). The catalyst layer **114** [i.e., of the gas diffusion electrode **111**] and the metal electrode **105** are abutted to each other with no gap therebetween, thus forming zero-gap configuration [best seen in FIGS. **2** and **3**]. In other words, configuration of the battery cell **100** having the electrolyte chamber **101** supporting the metal electrode **105** and the CO.sub.2 chamber **109** accommodating the gas diffusion electrode **111** aids in forming zero-gap when the metal electrode **105** and the gas diffusion electrode **111** abut to each other. The zero-gap configuration of the battery cell **100** minimizes ionic resistance by reducing distance that the ions is required to travel between the metal electrode **105** and the gas diffusion electrode **111**. Such configuration between the metal electrode **105** and the gas diffusion electrode **111** leads to faster ion transport and improves battery cell **100** reaction kinetics, thus results in improving performance of the battery cell **100**. Further, the electrochemical reaction at the metal electrode **105** and the gas diffusion electrode **111** occur efficiently as there is minimal loss of ionic conductivity, thus aids in achieving current density of greater than 120 mA cm 2. In

addition, ohmic losses caused due to resistance caused by the electrolyte is reduced, which contributes to power density greater than 40 mW cm 2 and open-circuit potential of 1.73 V of the battery cell **100**. Furthermore, the zero-gap configuration makes the battery cell **100** compact and lightweight. In an implementation, the battery cell **100** includes a second gasket **125** which is positioned to surround the gas diffusion electrode **111** and abut a portion of the inner surface **103** of the CO.sub.2 chamber **109**. The second gasket **125** forms a sealing between the gas diffusion electrode **111** and the second cavity **110** [i.e., the CO.sub.2 chamber **101**], forming a leak proof flow path for CO.sub.2.

(13) In an implementation, as seen in FIG. **1**, the battery cell **100** may include a membrane **115** made of material including, but not limiting to, a glass fibre. The membrane **115** may be positioned on the metal electrode **105** or on the second side **124** [thus, the catalyst layer **114**] of the gas diffusion electrode **111**, such that, the membrane **115** is sandwiched between the metal electrode **105** and the second side **124** of the gas diffusion electrode **111**. The membrane **115** prevents direct contact of the metal electrode **105** and the gas diffusion electrode **111**, and maintains electric neutrality by allowing ionic movement between metal electrode **105** and the gas diffusion electrode **111**. The membrane **115** facilitates flow of the electrolyte from the metal electrode **105** and ensures uniform electrolyte distribution between the metal electrode **105** and the first side **123** of the gas diffusion electrode **111**. In addition, the membrane **115** aids in preventing contamination of the catalyst layer **114** by ZnO formed during operation of the battery cell **100**.

(14) In an implementation, as seen in FIG. **4**, the gas diffusion electrode **111** may include an intermediate layer **116** which is sandwiched between the second side **124** of the gas diffusion electrode **111** (GDE) and the catalyst layer **114**. As an example, the intermediate layer **116** may be formed using but not limiting to a mixture of Polytetrafluoroethylene (PTFE) and carbon nanospheres. The intermediate layer **116** acts as a hydrophobic, water-repellent layer and creates localized CO.sub.2 environment. This prevents absorption of CO.sub.2 by the electrolyte at the first side **123** of the gas diffusion electrode **111** (GDE), thus mitigating crystallization on the first side **123**. Mitigating crystallization allows diffusion of CO.sub.2 from the CO.sub.2 chamber **109** towards the catalyst layer **114** through the first side **123** of the gas diffusion electrode **111** (GDE), thus resulting in efficient CO.sub.2 conversion during operation of the battery cell **100**.

(15) Referring again to FIGS. **1** and **2**, the battery cell **100** may include at least one shell **119** coupled to each of the electrolyte chamber **101** and the CO.sub.2 chamber **109**. The at least one shell **119** aids in mechanically connecting all the components of the battery cell **100** and keeps the battery cell **100** intact. In an implementation, the at least one shell **119**, the electrolyte chamber **101** and the CO.sub.2 chamber **109** are defined with a plurality of openings **120** to receive one or more connecting elements **121** for securing the at least one shell **119** at each of the electrolyte chamber **101** and the CO.sub.2 chamber **109** with each other, which in turn secures the components of the battery cell **100**. As an example, the one or more connecting elements **121** may be but not limiting to bolt-nut arrangement, screws and the like. The one or more connecting elements **121** aids in easy assembling and disassembling of the battery cell, thus aiding easy serviceability. Further, the battery cell **100** includes at least one electric terminal **122** extending from each of the metal electrode **105** and the gas diffusion electrode **111** for conducting electrons from the metal electrode **105** towards the gas diffusion electrode **111**.

(16) In an operational implementation of the battery cell **100** to generate, convert and/or distribute energy, initially CO.sub.2 may be allowed to flow into the CO.sub.2 chamber **109** through the second inlet **112**, and the electrolyte may be allowed to flow into the electrolyte chamber **101** through the first inlet **117**. As seen in FIG. **5**, the CO.sub.2 flowing through the CO.sub.2 chamber **109** diffuses through the first side **123** of gas diffusion electrode **111** (GDE) towards the second side **124** of the gas diffusion electrode **111** (GDE), and then exits through the second outlet **113** along with other gases formed during reaction at the gas diffusion electrode **111** (GDE). Subsequently, as best seen in FIG. **1** [indicated by arrow line], the electrolyte flowing through the

electrolyte chamber **101** passes through one aperture of the plurality of apertures **106** and penetrates through the membrane **115**. The membrane **115** aids uniform electrolyte distribution between the metal electrode **105** and the second side **123** of the gas diffusion electrode **111**, and then flows through other aperture of the plurality of apertures **106** and exits the electrolyte chamber **101** through the first outlet **104**, along with liquid products formed during the reaction.
(17) When CO.sub.2 flows through the CO.sub.2 chamber **109**, a chemical reaction of CO.sub.2 occurs at the gas diffusion electrode **111** (thus, cathode) as shown in equation (i), below:
CO.sub.2+2H.sub.2O+2e.fwdarw.HCOO+OH (i)
(18) That is, at the cathode, the CO.sub.2 supplied to the CO.sub.2 chamber **109** is subjected to chemical reaction with H.sub.2O in the electrolyte which is allowed to flow through the electrolyte chamber **101** and produce formate and hydroxide ions.
(19) Further, at the metal electrode **105** [Zn electrode], Zn is oxidized to form zinc ions (Zn.sup.2+), releasing electrons, as shown in equation (ii), below. Oxidation of Zn occurs electrochemically and is a spontaneous reaction driven by the thermodynamics of the system and the electrochemical potential difference established during the battery's operation.
Zn.fwdarw.Zn.sup.2+2e.sup.− (ii)
(20) Therefore, the reaction scheme of the overall reaction occurring in a reaction process during operation of the battery cell **100** is as shown in equation (iii), below
Zn+CO.sub.2+2H.sub.2O.fwdarw.Zn(OH).sub.2+HCOO (iii)
(21) In addition to above, additional chemical reactions occur in the electrolyte, where Hydroxide is formed as shown in equation (iv) and precipitation of zinc hydroxide occurs as shown in equation (v), as below
H.sub.2O+e.sup.−.fwdarw.OH.sup.−+12H.sub.2 (iv)
Zn.sup.2+2OH.sup.−.fwdarw.Zn(OH).sub.2 (v)
(22) In conclusion and as depicted in the above chemical reactions, CO.sub.2 flows through the gas diffusion electrode **111** CO.sub.2 reacts with protons (H) and electrons (e) to form various products based on the catalyst selectivity, such as CO, formate (HCOO), methane (CH.sub.4), or ethylene (C.sub.2H.sub.4). Concurrently in alkaline conditions, hydroxide ions (OH) are formed at the gas diffusion electrode **111**. The generated hydroxide ions (OH) migrate through the electrolyte toward the metal electrode **105**. At the metal electrode **105** (i.e., Anode side) zinc (Zn) is oxidized by reacting with hydroxide ions (OH), forming zinc hydroxide (Zn(OH).sub.2) and releasing electrons (e). Zinc hydroxide (Zn(OH).sub.2) further decomposes into zinc oxide (ZnO) and water. The ZnO deposits on the metal electrode **105**. The electrons released during the zinc oxidation reaction at the metal electrode **105** flow through the external circuit (load), generating an electric current that powers the connected external load. The electrons then return to the gas diffusion electrode **111**, completing the circuit and allowing the chain of reactions to continue.
(23) It should be noted that, the chemical reaction occurred at the gas diffusion electrode **111**, the metal electrode **105** and additional reactions depends on metal type of the metal electrode **105**, catalyst material, electrolyte composition and battery cell **100** operating conditions, and the same should not be construed as an operating limitation of the battery cell **100** of the present disclosure. As an example, if Copper-based catalysts are used in the battery cell, hydrocarbons such as CH.sub.4, C.sub.2H.sub.4 is obtained. While other catalysts lead to formation of CO, HCOO, or CH.sub.3OH, which may be subjected to secondary processes to obtain hydrocarbons.
(24) Example
(25) The zero-gap flow type metal-CO.sub.2 battery cell of the present disclosure was subjected to open-circuit potential (OCP) examination under continuous supply of CO.sub.2. Polarization curves were obtained for charging and discharging cycles of the zero-gap flow-type metal-CO.sub.2 battery cell. Electrolyte flow rate was maintained at 10 mL min.sup.−1 and CO.sub.2 gas was introduced at a flow rate of 40 sccm (standard cubic centimeters per minute). Polarization curves obtained have been depicted in FIGS. **7***a* to **7***c*.

(26) From FIG. **7***a*, when the atmosphere was switched to Argon, OCP fluctuations were observed, indicating the CO_2 reduction capability of the zero-gap flow-type metal-CO_2 battery cell battery cell. Further, FIG. **7***b* depicts the discharge and charge polarization curves, along with the corresponding power density curve. These curves demonstrate a power density of 43 mW cm″2 and a current density of 120 mA cm″2-values significantly higher than those reported for conventional Zn—CO_2 batteries (<7 mW cm″2 and <15 mA cm″2).
(27) To assess long-term stability, the battery was subjected to continuous discharge for 23 hours at a constant current rate of 0.35 mA cm″2, as depicted in FIG. **7***c*. During this period, the battery displayed excellent stability, maintaining a high discharge voltage of 1.3 V, with only a minimal voltage decrease over the 23-hour discharge.
(28) Additionally, the zero-gap flow-type metal-CO_2 battery cell exhibited high performance, particularly during the discharge process, achieving a faradaic efficiency (FE) of 96.3% for formate ($HCOO^-$) production. This high FE underscores the battery's strong capability to efficiently convert CO_2 into electricity and valuable chemical products. These results highlight the potential of the zero-gap flow-type metal-CO_2 battery cell as a stable and efficient energy conversion system for CO_2 utilization. It should be imperative that the device, the system, and any other elements described in the above description should not be considered as a limitation with respect to the figures. Rather, variations to such devices, systems and methods should be considered within the scope of the description.

Claims

1. A metal-Carbon dioxide (CO_2) battery cell, the battery cell comprising: an electrolyte chamber configured to allow flow of electrolyte; a metal electrode configured as an anode, the metal electrode is accommodated in the electrolyte chamber and is defined with a plurality of apertures; a Carbon dioxide (CO_2) chamber configured to allow flow of Carbon dioxide (CO_2); and a gas diffusion electrode (GDE) configured as a cathode, the gas diffusion electrode (GDE) is accommodated in the Carbon dioxide (CO_2) chamber, the gas diffusion electrode (GDE) is defined with a first side and a second side, opposite to the first side, and the second side is coated with a catalyst layer, wherein, the catalyst layer abuts a membrane on the metal electrode such that, the electrolyte from the electrolyte chamber flows through the plurality of apertures in the metal electrode, penetrates through the membrane and contacts the catalyst layer, and Carbon dioxide (CO_2) from the Carbon dioxide (CO_2) chamber contacts the second side of the gas diffusion electrode to diffuse through the catalyst layer towards the metal electrode, and wherein the membrane is made of a glass fibre.
2. The battery cell according to claim 1, wherein the gas diffusion electrode comprises an intermediate layer sandwiched between the first side of the gas diffusion electrode and the catalyst layer, wherein the catalyst layer is formed using a mixture of Polytetrafluoroethylene (PTFE) and carbon nanospheres.
3. The battery cell according to claim 1, wherein the metal electrode is a Zinc electrode.
4. The battery cell according to claim 1, wherein the gas diffusion electrode is made of carbon paper.
5. The battery cell according to claim 1, wherein the electrolyte is 6M KOH+0.02M $Zn(CH_3COO)_2$).
6. The battery cell according to claim 1, wherein the catalyst layer is formed of Bismuth metal-organic framework (Bi-MOF).
7. The battery cell according to claim 1, wherein the electrolyte chamber is defined with a first inlet and a first outlet, wherein the electrolyte enters through the first inlet and discharges through the first outlet.
8. The battery cell according to claim 1, wherein the electrolyte chamber is defined with a first

cavity on an inner surface and extends along a periphery at a distance from sides of the electrolyte chamber.

9. The battery cell according to claim 8, comprising a first gasket surrounding the metal electrode and abutting to inner walls of the first cavity.

10. The battery cell according to claim 1, wherein the Carbon dioxide (CO.sub.2) chamber is defined with a second inlet and a second outlet, wherein Carbon dioxide (CO.sub.2) enters through the second inlet and discharges through the second outlet.

11. The battery cell according to claim 1, wherein the Carbon dioxide (CO.sub.2) chamber is defined with a second cavity on an inner surface and extends along a periphery at a distance from sides of the Carbon dioxide (CO.sub.2) chamber.

12. The battery cell according to claim 11, comprising a second gasket surrounding the gas diffusion electrode and abutting inner walls of the second cavity.

13. The battery cell according to claim 1, comprising at least one electric terminal extending from each of the metal electrode and the gas diffusion electrode to conduct electrons from the metal electrode towards the gas diffusion electrode.

14. A metal-Carbon dioxide battery module, the module comprising: a casing; and a plurality of metal-Carbon dioxide battery cells stacked within the casing in a defined orientation, each of the plurality of metal-Carbon dioxide battery cells comprises: an electrolyte chamber configured to allow flow of electrolyte; a metal electrode configured as an anode, the metal electrode is accommodated in the electrolyte chamber and is defined with a plurality of apertures; a Carbon dioxide (CO.sub.2) chamber configured to allow flow of Carbon dioxide (CO.sub.2); and a gas diffusion electrode (GDE) configured as a cathode, the gas diffusion electrode (GDE) is accommodated in the Carbon dioxide (CO.sub.2) chamber, the gas diffusion electrode (GDE) is defined with a first side and a second side, opposite to the first side, and the second side is coated with a catalyst layer, wherein, the catalyst layer abuts a membrane on the metal electrode such that, the electrolyte from the electrolyte chamber flowing through the plurality of apertures in the metal electrode penetrates through the membrane and contacts the catalyst layer, and Carbon dioxide (CO.sub.2) from the Carbon dioxide (CO.sub.2) chamber contacts the second side of the gas diffusion electrode to diffuse through the catalyst layer towards the metal electrode, and wherein the membrane is made of a glass fibre.

15. The battery module according to claim 14, comprising a pump adapted to circulate the electrolyte and Carbon dioxide to each of the plurality of metal-Carbon dioxide battery cells stacked within the casing.

16. The battery cell according to claim 7, wherein the electrolyte chamber is defined with a first cavity on an inner surface and extends along a periphery at a distance from sides of the electrolyte chamber.

17. The battery cell according to claim 8, comprising a first gasket surrounding the metal electrode and abutting to inner walls of the first cavity.

18. The battery cell according to claim 16, comprising a first gasket surrounding the metal electrode and abutting to inner walls of the first cavity.

19. The battery cell according to claim 11, comprising a second gasket surrounding the gas diffusion electrode and abutting inner walls of the second cavity.